Expanded Three-Player Bilevel Benchmark

Notation

There are three players. Player p's decision vector is

$$x_p = (x_{p,1}, x_{p,2}, x_{p,3}, x_{p,4}, x_{p,5}) \in \mathbb{R}^5.$$

The bilevel structure used in the tests is:

$$\text{upper level controls coordinates } 4, 5, \qquad \text{lower level controls coordinates } 1, 2, 3.$$

For each player problem below, the other players' decisions are treated as fixed parameters, as in a generalized Nash equilibrium. The three player problems are solved simultaneously.

Player 1

Player 1 solves

$$\begin{aligned} \min_{x_1 \in \mathbb{R}^5} \quad & (x_{1,4} - 1.20)^2 + (x_{1,5} - 0.80)^2 \\ \text{subject to} \quad & x_1 \in \mathcal{S}_1(x_2, x_3). \end{aligned}$$

The lower-level solution set is

$$\begin{aligned} \mathcal{S}_1(x_2, x_3) = \arg\min_{y_1 \in \mathbb{R}^5} \quad & (y_{1,1} - 1.00)^2 + (y_{1,2} - 0.90)^2 + (y_{1,3} - 0.15)^2 \\ \text{subject to} \quad & 0.6375 - y_{1,1} - 0.25x_{2,1} - 0.25x_{3,1} \geq 0, \\ & 2.4250 - y_{1,2} - 0.25x_{2,2} - 0.25x_{3,2} \geq 0, \\ & y_{1,3} + 0.25x_{2,3} + 0.25x_{3,3} - 0.8625 \geq 0, \\ & y_{1,4} + 0.25x_{2,4} + 0.25x_{3,4} - 0.8750 \geq 0. \end{aligned}$$

Player 2

Player 2 solves

$$\begin{aligned} \min_{x_2 \in \mathbb{R}^5} \quad & (x_{2,4} - 1.30)^2 + (x_{2,5} - 0.90)^2 \\ \text{subject to} \quad & x_2 \in \mathcal{S}_2(x_1, x_3). \end{aligned}$$

The lower-level solution set is

$$\begin{aligned} \mathcal{S}_2(x_1, x_3) = \arg\min_{y_2 \in \mathbb{R}^5} \quad & (y_{2,1} - 1.05)^2 + (y_{2,2} - 1.00)^2 + (y_{2,3} - 0.20)^2 \\ \text{subject to} \quad & 0.6750 - y_{2,1} - 0.25x_{1,1} - 0.25x_{3,1} \geq 0, \\ & 2.5000 - y_{2,2} - 0.25x_{1,2} - 0.25x_{3,2} \geq 0, \\ & y_{2,3} + 0.25x_{1,3} + 0.25x_{3,3} - 0.9000 \geq 0, \\ & y_{2,4} + 0.25x_{1,4} + 0.25x_{3,4} - 0.9500 \geq 0. \end{aligned}$$

Player 3

Player 3 solves

$$\begin{aligned} \min_{x_3 \in \mathbb{R}^5} \quad & (x_{3,4} - 1.40)^2 + (x_{3,5} - 1.00)^2 \\ \text{subject to} \quad & x_3 \in \mathcal{S}_3(x_1, x_2). \end{aligned}$$

The lower-level solution set is

$$\begin{aligned} \mathcal{S}_3(x_1, x_2) = \arg\min_{y_3 \in \mathbb{R}^5} \quad & (y_{3,1} - 1.10)^2 + (y_{3,2} - 1.10)^2 + (y_{3,3} - 0.25)^2 \\ \text{subject to} \quad & 0.7125 - y_{3,1} - 0.25x_{1,1} - 0.25x_{2,1} \geq 0, \\ & 2.5750 - y_{3,2} - 0.25x_{1,2} - 0.25x_{2,2} \geq 0, \\ & y_{3,3} + 0.25x_{1,3} + 0.25x_{2,3} - 0.9375 \geq 0, \\ & y_{3,4} + 0.25x_{1,4} + 0.25x_{2,4} - 1.0250 \geq 0. \end{aligned}$$

Expected Equilibrium

The expected equilibrium used in the tests is

$$\begin{aligned} x_1^\star &= (0.40,\ 0.90,\ 0.55,\ 1.20,\ 0.80), \\ x_2^\star &= (0.45,\ 1.00,\ 0.60,\ 1.30,\ 0.90), \\ x_3^\star &= (0.50,\ 1.10,\ 0.65,\ 1.40,\ 1.00). \end{aligned}$$

At this point, each player's first and third constraints are active, and each player's second and fourth constraints have slack equal to 1. For example, for Player 1,

$$\begin{aligned} 0.6375 - 0.40 - 0.25(0.45) - 0.25(0.50) &= 0, \\ 2.4250 - 0.90 - 0.25(1.00) - 0.25(1.10) &= 1, \\ 0.55 + 0.25(0.60) + 0.25(0.65) - 0.8625 &= 0, \\ 1.20 + 0.25(1.30) + 0.25(1.40) - 0.8750 &= 1. \end{aligned}$$

Nonlinear Test Variant

The nonlinear benchmark in the tests uses the same expanded player problems and the same expected equilibrium, but replaces every squared-distance term

$$(z - a)^2$$

by

$$\cosh(z - a) - 1,$$

and replaces every linear inequality residual $r(z) \geq 0$ by

$$\exp(r(z)) - 1 \geq 0.$$

This preserves the same active and inactive constraints.